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Original Article

# From vulnerability to robustness: Radiation-hard isolation for BPR-enabled stacked nanosheet CFETs

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ABSTRACT

The integration of Buried Power Rail (BPR) and Complementary FET (CFET) technologies is a promising way to improve power efficiency and circuit density in advanced logic devices. However, the radiation reliability of BPR-integrated logic and memory circuits remains insufficiently explored. For the first time, we provide a quantitative analysis of Single Event Effects (SEE) in BPR-integrated Si-CFET structures using semiconductor device- and circuit-level simulator. Simulation results show that the BPR structure increases the local electric field intensity by approximately 20.2 % compared to the conventional Front Power Rail (FPR), leading to extended charge collection paths and transient current densities up to 41.5 % higher under identical irradiation conditions. Moreover, mixed-mode simulations of SRAM latch circuits indicate that bit-flips can occur at Linear Energy Transfer (LET) levels as low as 1–2 MeV cm$^2$/mg, indicating significant vulnerability under low-voltage operation. To address this, structural mitigation strategies—such as Buried Dielectric Isolation and Backside Contact—are proposed to increase the threshold LET margin to 34 MeV cm$^2$/mg, representing a 17-fold improvement, without degrading device performance. These results offer critical design insights for enhancing radiation robustness in ultra-scaled logic and memory architectures using BPR integration.

## 1. Introduction

With the industry now entering the sub-3 nm era, conventional FETs face fundamental limits, making alternative device architectures indispensable [1–3]. Logic devices have evolved from planar MOSFETs to FinFETs and, more recently, to Gate-All-Around (GAA) FETs, which offer improved electrostatic control and scalability [4–6]. Among the latest developments, the vertically stacked Complementary FET (CFET) architecture has emerged as a promising candidate for next-generation logic applications [7–10]. By vertically integrating n-type and p-type MOSFETs, the CFET significantly reduces the device footprint and improves power, performance, and area (PPA) metrics [11,12].

However, as device scaling continues, longer interconnects and thinner upper-level wires have become critical bottlenecks, causing power loss and performance degradation [13–16]. Because the conventional Front Power Rail (FPR) shares its routing plane with signal lines in the top metal layers, it exacerbates congestion and raises path resistance. This configuration leads to larger IR drops and diminished power delivery efficiency [17–19]. As a result, FPR-based architecture is no longer sufficient to meet the power integrity requirements of aggressively scaled technology nodes and has become a limiting factor in system-level performance. To address these challenges, both industry and academia have recently introduced the Buried Power Rail (BPR) structure as a novel interconnect paradigm that relocates power rails beneath the active device layer, thereby offering enhanced power delivery and layout flexibility [20,21].

The BPR architecture significantly shortens the power delivery path, thereby reducing parasitic resistance and capacitance (RC) and mitigating IR drop [21]. By relocating power rails beneath the device layer, BPR also alleviates routing congestion and enhances design flexibility.

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In particular, the integration of BPR with CFET architecture creates a synergistic effect, combining the area reduction benefits of vertical stacking with the RC minimization enabled by shortened power paths. The vertical stacking nature of CFET combined with the shortened power routing of BPR achieves superior area efficiency, power integrity, and thermal management, making this integration highly advantageous for next-generation ultra-scaled semiconductor technologies. This integration is essential for meeting the stringent performance and energy efficiency demands of highly scaled technology nodes [22–24].

Accordingly, BPR integration is now essential for power delivery in advanced nodes [25] and is poised to expand into high-reliability domains such as space systems. In particular, various logic devices, including memory components such as SRAM must ensure robustness against radiation-induced disturbances when deployed in such harsh conditions [26–28]. Accordingly, a systematic and comprehensive evaluation of radiation hardness in BPR-integrated device architectures is imperative to ensure reliable operation under space and radiation-prone environments.

While FinFET-based radiation-hardened logic has already been adopted and qualified by aerospace agencies [51], there are currently no reports of CFET technology being applied in operational space systems. Nevertheless, as semiconductor scaling advances beyond the FinFET generation, GAA-FETs and vertically stacked CFETs are widely regarded as promising successors for post-3 nm logic nodes due to their superior electrostatic control, reduced power consumption, and enhanced integration density. Consequently, evaluating the radiation response of these emerging device architectures, including CFETs, is a necessary step toward enabling next-generation, high-density, and energy-efficient semiconductor platforms suitable for aerospace and high-reliability applications [37,55].

With the advancement of next-generation FET architectures, ensuring radiation reliability under extreme environmental conditions has become an increasingly critical aspect of logic device design. Applications in aerospace, automotive electronics, and high-performance computing frequently expose semiconductor devices to high-energy particles such as protons, heavy ions, alpha particles, and neutrons, originating from cosmic rays, solar flares, and radioactive isotopes within packaging materials [29]. These particles interact with semiconductor materials, resulting in various radiation-induced effects that are generally classified into three primary mechanisms: Displacement Damage (DD), Total Ionizing Dose (TID), and Single Event Effects (SEE) [30–32].

Among these mechanisms, SEEs are transient phenomena triggered by the instantaneous interaction between a single high-energy particle and a semiconductor device [32]. Unlike cumulative damage mechanisms, SEE can result in non-recoverable logic malfunctions or transient current pulses. When a high-energy particle penetrates the silicon substrate, it generates a dense track of electron-hole pairs. These charge carriers are rapidly driven by internal electric fields, producing a transient current known as a Single Event Transient (SET). Such events can lead to logic state disturbances, memory bit flips (soft errors), and, in severe cases, permanent hardware damage (hard errors).

As devices continue to scale below several tens of nanometers, the reduction in gate length leads to a corresponding decrease in storage node capacitance and a lower critical charge ($Q_{crit}$), thereby increasing the sensitivity of devices to SEEs [33–35]. Consequently, modern logic circuits, characterized by high integration density and low-power operation, are inherently vulnerable to radiation-induced transient events. Accordingly, a thorough understanding of SEE mechanisms, along with the implementation of effective structural countermeasures, is essential for maintaining system-level reliability.

Previous radiation-hardness studies have mostly quantified SET/SEU behavior, $Q_{crit}$, and soft-error rates in FinFET, nanosheet-FET, and CFET architectures via TCAD and circuit-level simulations [36,37]. In parallel, machine learning-based data analysis has gained increasing attention as a complementary approach, enabling fast prediction of radiation responses and charge collection behavior, as well as facilitating AI-driven design space exploration for radiation-hardened architectures [38]. However, these studies have largely focused on comparative sensitivity assessments across different architectures, with limited investigation into the intrinsic radiation-response mechanisms of CFETs or the structural implications of BPR integration. Unlike previous studies, this work specifically identifies and quantifies the impact of enhanced electric fields induced by BPR integration on radiation-induced charge collection dynamics, offering novel insights and practical guidelines for designing radiation-hardened BPR-CFET devices. In particular, the enhanced electric field induced by the proximity between the BPR metal and the silicon substrate has not been sufficiently investigated from a computational or physical standpoint in terms of its effect on charge collection efficiency.

This paper is organized as follows: Section II describes the simulation environment and device models used in this study. Section III presents a detailed analysis of electric-field distributions, charge collection dynamics, and SEE sensitivities in BPR-integrated CFETs. Section IV discusses effective structural mitigation techniques to enhance radiation hardness, and finally, Section V summarizes the findings in this work.

## 2. Simulation environment

We first built a three-dimensional BPR-CFET structure and then added the electrical and radiation-physics models described below.

In this study, the radiation tolerance of a three-dimensional CFET structure incorporating a BPR was evaluated through an integrated simulation approach based on the process flow illustrated in Fig. 1. As illustrated in Fig. 1, the simulation process begins with the definition and calibration of the device geometry, followed by configuring the radiation model parameters and bias conditions. Subsequently, TCAD device-level simulations are performed to analyze transient phenomena, and finally, circuit-level simulations are executed to evaluate the practical implications of radiation-induced effects. The device geometry was defined and calibrated using measured data to ensure simulation accuracy and electrical model validity. Subsequently, radiation particle parameters and biasing conditions were configured to reflect realistic operating environments. Using 3D TCAD simulations, we systematically analyzed electron-hole pair generation induced by high-energy particle strikes, electric field and current density distributions, transient drain current responses, and total collected charge.

The device-level simulation results were further integrated into circuit-level analyses to evaluate radiation-induced reliability impacts in practical circuit environments, such as inverters and SRAM cells. This allowed for a comprehensive assessment of output voltage perturbations and potential bit-flip occurrences under radiation exposure. Through this multi-scale evaluation framework, the charge collection dynamics and circuit-level failure mechanisms in BPR-CFET structures were precisely characterized. Based on these insights, this study proposes practical design guidelines to enhance the radiation robustness of next-generation semiconductor devices.

### 2.1. Device modeling and electrical characteristics calibration

In this work, a monolithic CFET inverter incorporating BPR was implemented using a vertically stacked NMOS-on-PMOS silicon-based Nanosheet FET (NS-FET) architecture. Three-dimensional electrical simulations were performed using Synopsys Sentaurus TCAD.

Fig. 2 shows a schematic process flow for the Monolithic CFET. The device is fabricated through sequential epitaxial growth of Si/SiGe multilayers, selective etching of sacrificial layers for channel release, and subsequent gate stack and contact formation. This process enables true vertical stacking of n- and p-type nanosheet channels within a single wafer, realizing a self-aligned and highly compact complementary configuration. Compared with sequentially bonded CFETs, the monolithic approach offers superior alignment accuracy, reduced parasitic

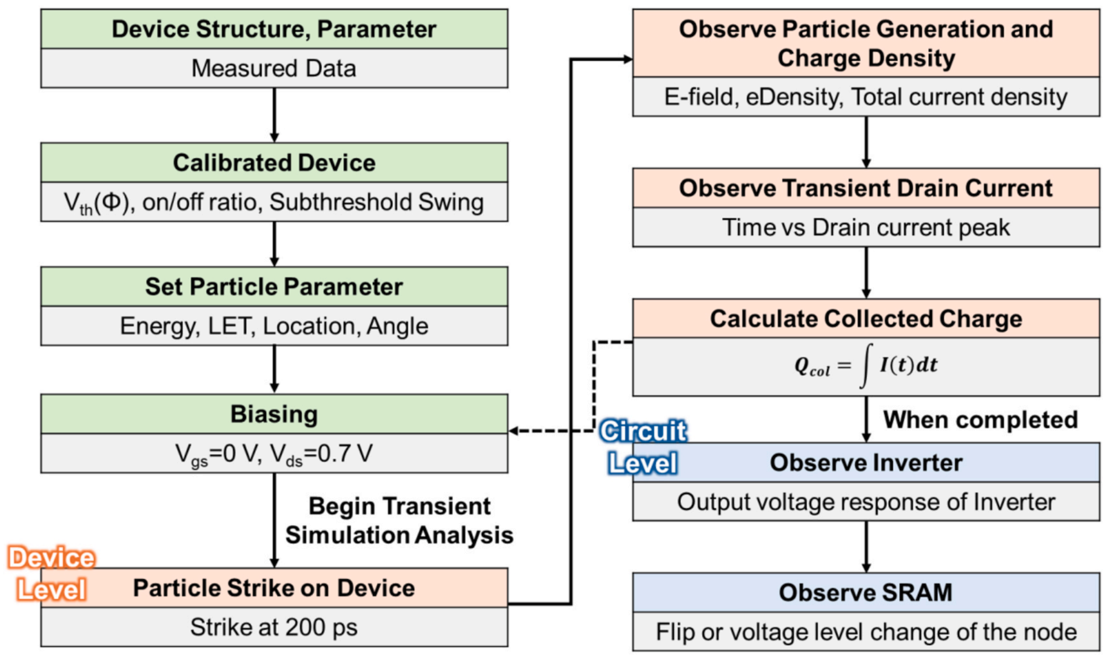


**Fig. 1.** Process flow for device- and circuit-level Single Event Effect evaluation using TCAD simulation.

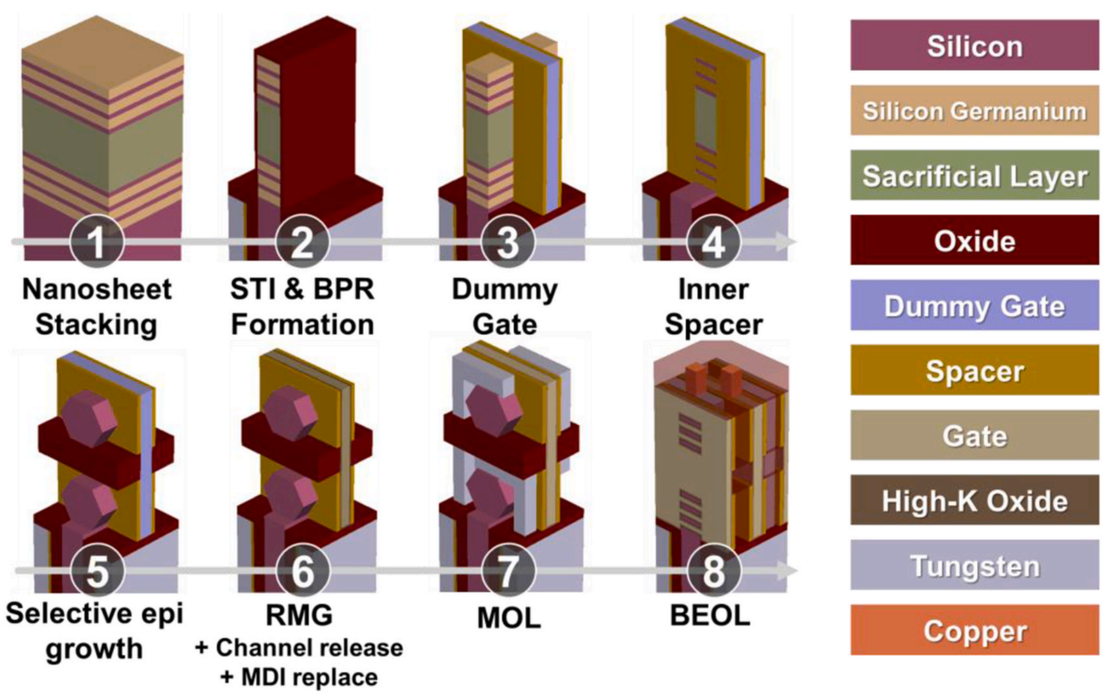


**Fig. 2.** Fabrication flow of the Monolithic Si Complementary FET.

coupling due to the absence of bonding dielectrics, and higher integration density.

Fig. 3(a)–(c) depict the overall device structure and key cross-sectional views. The source/drain contacts for the NMOS and PMOS transistors were designed independently, while the vertically stacked N/P channels were simultaneously controlled through a shared gate electrode. To ensure optimal device performance, distinct metal work functions were applied to the NMOS and PMOS channels, enabling proper threshold voltage alignment and electrical symmetry.

The doping profiles consisted of a channel region doped to $1 \times 10^{15}$ $cm^{-3}$ and source/drain regions doped to $5 \times 10^{20}$ $cm^{-3}$. A punch-through stopper (PTS) implant was introduced into the substrate using a Gaussian distribution with a peak concentration of $5 \times 10^{18}$ $cm^{-3}$ [39]. Each terminal node was assigned a contact resistivity of $1 \times 10^{-9}$ $\Omega$ $cm^2$ [40,41].

For the middle-of-line (MOL) local interconnects, including the M0 layer and VIA structures, tungsten (W) with a 2 nm titanium nitride (TiN) barrier was assumed as the metal material [42,43]. The width of the BPR was designed to be approximately 1.5 × the M0 metal pitch [42,44]. The resistivity and contact resistivity values for the M0 and VIA metals were assigned based on prior experimental data and literature sources [45,46]. Key physical dimensions, such as contact poly pitch

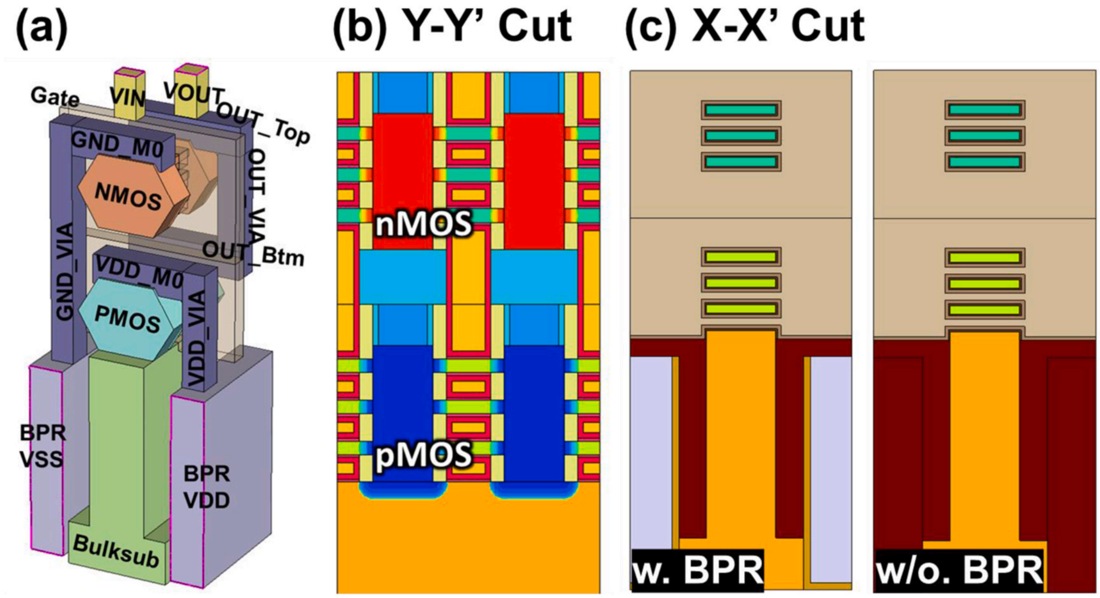


**Fig. 3. Device model of Si Complementary FET** (a) 3D Schematic view of the 3D CFET structure (b), (c) Cross-sectional views (X-X′, Y-Y′) of the CFET structure.

(CPP) and the spacing between NMOS and PMOS devices, were determined with reference to recent fabrication-oriented studies [47–49]. A detailed summary of these parameters is provided in Table I.

To ensure the reliability of device simulations, front-end-of-line (FEOL) parameters for the NS-FET were configured based on experimentally characterized device structures [50], followed by iterative calibration against the measured transfer characteristics ($I_d$-$V_g$) of a 1.4 nm node NS-FET, as shown in Fig. 4. Calibration was performed under both linear ($V_{DS} = 0.05$ V) and saturation ($V_{DS} = 0.7$ V) bias conditions. The junction abruptness was set to 4 nm/decade, and the peak doping concentration of the source/drain extension was adjusted to $5 \times 10^{20}$ $cm^{-3}$ to match subthreshold swing and Drain-Induced Barrier Lowering (DIBL) characteristics. Furthermore, low-field mobility models were tuned in the linear regime, while high-field saturation models were adjusted in the saturation regime to accurately reproduce the experimental transfer curves.

All device simulations were systematically calibrated to reflect actual operating conditions for each structure, ensuring high simulation accuracy. The drift diffusion-based carrier transport model was carefully tuned to reproduce both linear and saturation I-V characteristics through direct comparison with measured data. To accurately model

**Table 1**
Key Device Parameters used in this work.

| Dimension Parameter | Value |
|---|---|
| Contacted Poly Pitch (CPP) | 48 nm |
| Metal Pitch | 28 nm |
| Gate Length | 12 nm |
| Outer and Inner Spacer Length | 5 nm |
| Contact Length | 16 nm |
| Nanosheet Thickness | 5 nm |
| Nanosheet-to-Nanosheet Spacing (NS-to-NS) | 10 nm |
| Nanosheet Width | 30 nm |
| Number of Nanosheets | 3 |
| S/D Side Epi Width at Nanosheet | 15 nm |
| Interfacial Layer Thickness | 1 nm |
| High-k ($HfO_2$) Thickness | 2 nm |
| N/P Separation | 30 nm |
| Source/Drain Doping Concentration | $5 \times 10^{20} cm^{-3}$ |
| Bottom Doping Concentration | $2 \times 10^{18} cm^{-3}$ |
| VIA Width | 16 nm |
| BPR Width | 42 nm |
| Contact Resistivity | $1 \times 10^{-9}$ $\Omega$*$cm^{-2}$ |
| M0 Resistivity | $3.9 \times 10^{-5}$ $\Omega$*$cm^{-2}$ |

carrier behavior under radiation exposure, Shockley Read Hall (SRH) and Auger recombination models, as well as the impact ionization model, were activated. Furthermore, the combined use of the impact ionization, SRH/Auger recombination, and substrate potential coupling models inherently reflect secondary mechanisms such as the parasitic bipolar amplification effect (PBAE) during transient radiation events. Carrier mobility degradation due to phonon and Coulomb scattering in the silicon bulk was modeled using the Philips unified mobility model, while thin-layer mobility degradation resulting from the ultra-scaled channel thickness of approximately 10 nm was accounted for by incorporating the thin layer mobility model. For high electric field conditions, both the high field saturation model and the velocity saturation model were employed. Under low field conditions, ballistic transport and the Lombardi mobility model were integrated to capture field-dependent mobility behavior. Additionally, Coulomb and surface roughness scattering models were included to quantitatively account for scattering effects induced by high doping concentrations and degraded interface quality.

The selected physical models were calibrated to ensure both the physical validity and quantitative accuracy of the simulation. Each model was carefully adjusted to match the measured current and voltage characteristics of actual devices. Key model parameters were iteratively tuned based on the device operating region, either linear or saturation, as well as electric field strength, doping concentration, and channel thickness. Through this process, critical electrical metrics such as subthreshold swing, saturation current, and DIBL were accurately reproduced and closely aligned with experimental data. This calibration procedure serves as the fundamental basis for ensuring the reliability of subsequent structure-dependent comparisons and radiation induced-reliability assessments.

## 2.2. Radiation modeling

The generation of electron-hole pairs under radiation exposure is primarily induced by alpha particles and heavy ions. Alpha particles, which are helium nuclei emitted from the radioactive decay of impurities within the packaging materials, possess relatively low energy and shallow penetration depth. As a result, they create localized charge tracks near the silicon surface, often leading to SETs. In contrast, heavy ions exhibit significantly greater mass and energy, depositing high LET across the entire device volume. This can lead to more severe Single-

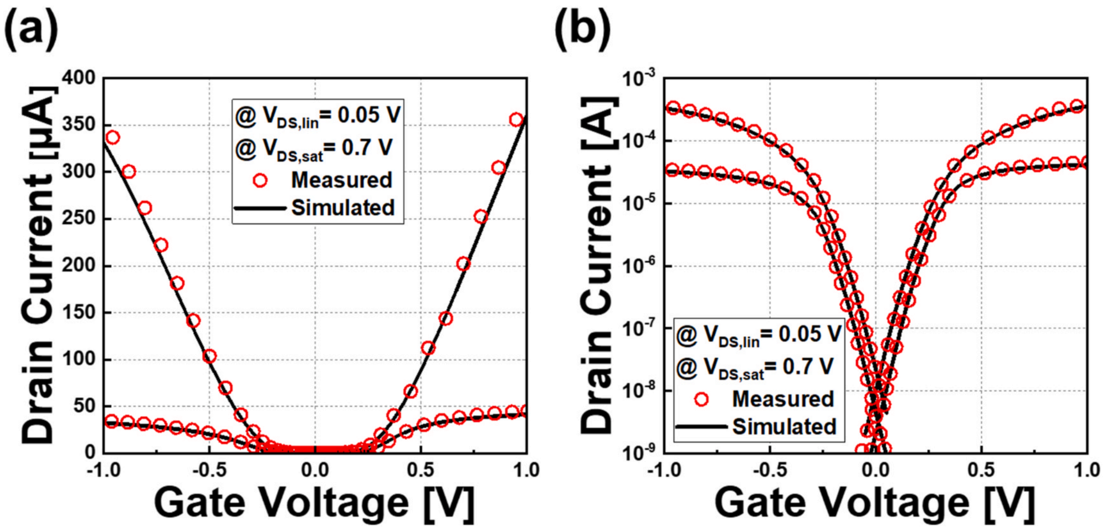


**Fig. 4.** Calibration results for transfer curves of N/P Nanosheet FETs in (a) linear scale and (b) log scale.

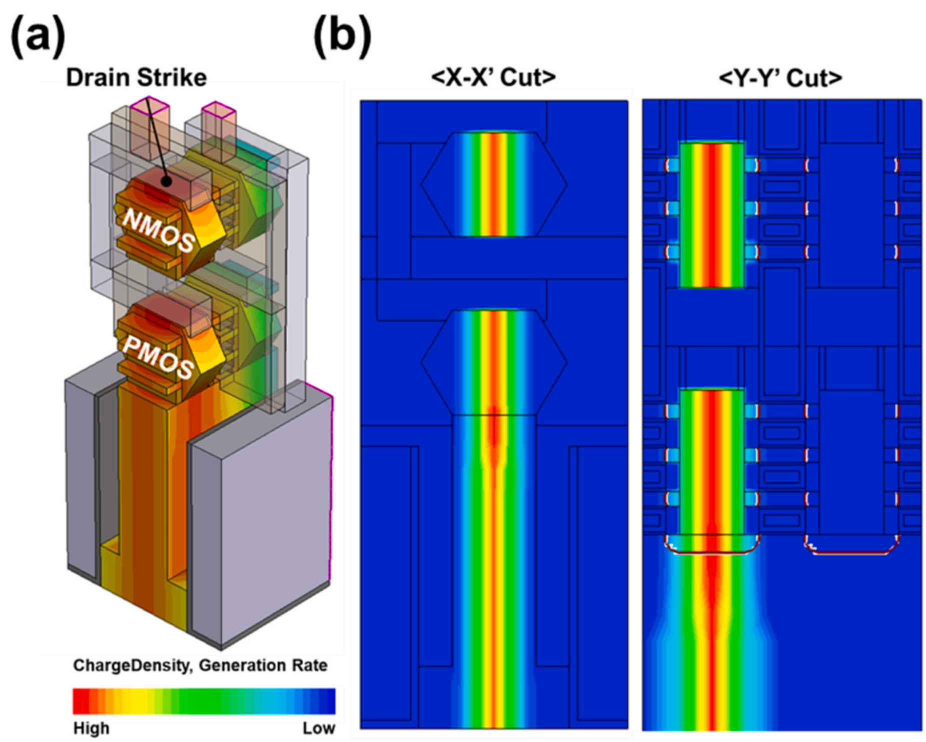


**Fig. 5. Radiation-induced particle profile for Si-CFET structures** (a) 3D view of a Si-CFET, illustrating the drain strike location and charge density distribution. (b) Cross-sectional views of radiation generation profiles in the Si-CFET, showing temporal evolution after a drain strike.

Event Effects, including bit flips and, in extreme cases, latch-up.

For SEE simulations, a worst-case scenario was assumed by setting the particle incident angle to 90° and targeting the strike location at the drain region [51] (see Fig. 5). A vertical incidence angle of 90° at the drain node represents a worst-case scenario, as this condition maximizes localized charge collection efficiency and transient current magnitudes due to the shortest possible path to the sensitive junction regions. The drain is known to be the most charge-sensitive node within the device, where charge collection efficiency is maximized. When a high-energy particle vertically strikes this region, the generated electron-hole pairs are concentrated in a confined volume, resulting in a high charge generation rate per unit time and volume. Consequently, the collected charge is directly funneled into the drain terminal, increasing the likelihood of transient current pulses compared to strikes in other locations.

The alpha particle and heavy ion radiation models employed in this study were based on the default implementations provided by the Sentaurus TCAD. The electron-hole pair generation rate induced by alpha particles, denoted as G(u,v,w,t), was modeled using a physics-based formulation and implemented within the TCAD simulation framework (Eq. (1)) [52,53].

$$G(u,v,w,t) = \frac{a}{\sqrt{2\pi \cdot s}} \exp\left[-\frac{1}{2}\left(\frac{t-t_m}{s}\right)^2 - \frac{1}{2}\left(\frac{v^2+w^2}{w_t^2}\right)\right] \left[c_1 e^{\alpha u} + c_2 \exp\left(-\frac{1}{2}\left(\frac{u-\alpha_1}{\alpha_2}\right)^2\right)\right] \tag{1}$$

In this model, u, v and w represent the orthogonal coordinates along the ion track path. The parameter $t_m$ [s] denotes the temporal peak of the Gaussian generation profile, while s defines the temporal standard deviation. The term $w_t$ [nm] corresponds to the radial spread of the Gaussian EHP distribution. The coefficients a, $C_1$, $C_2$, $a_1$, and $a_2$ are constant parameters, where $a_1$ and $C_1$ are functions of the particle energy E and are defined as follows. (Eq. (2)) and (Eq. (3)) [53].

$$a_1 = a_0 + a_1 E + a_2 E^2 \tag{2}$$

$$c1 = \exp\left[a(a_1[10MeV] - a_1[E])\right] \tag{3}$$

The scaling factor a is determined according to the following equation (Eq. (4)).

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} G(u,v,w,t)dtdwdvdu = \frac{E}{E_p} \tag{4}$$

In this equation, $E_p$ [eV] represents the average energy required to generate an electron-hole pair. Parameter values were taken from the default silicon deck in Sentaurus TCAD and used without modification [52]. A summary of the alpha particle parameters used is provided in Table II.

As heavy ions traverse the device structure, they lose energy through ionization processes, resulting in the formation of dense electron-hole pair tracks along their path. The generated EHPs can induce transient currents of sufficient magnitude to alter the logic state of the device, potentially leading to Single Event Upset (SEU). Unlike alpha particles, which are relatively slow and deposit energy over a localized region, heavy ions possess higher mass and velocity, enabling deeper penetration and broader charge deposition profiles. The carrier generation rate induced by heavy ion irradiation is modeled using the following expression (Eq. (5)). [52].

$$G(l,w,t) = G_{LET}(l)R(w.l)T(t) \tag{5}$$

R(w,l) represents the spatial distribution of generated carriers as a function of the perpendicular distance w from the ion track axis at a given position l. This distribution can be modeled using either an exponential or Gaussian function. In this study, a Gaussian profile is adopted, where w(l) denotes the radial characteristic distance from the ion track. The temporal variation of carrier generation is described by T (t), which is defined as a Gaussian function with respect to time, capturing the transient nature of particle-induced ionization events (Eq. (6)). and (Eq. (7)) [52].

$$R(w,l) = \exp\left(-\left(\frac{w}{wt(l)}\right)^2\right) \tag{6}$$

**Table 2**
Alpha particle parameters used in this work.

| Parameter name | Value |
|---|---|
| s | $2 \times 10^{-12}$ [s] |
| $c_2$ | 1.4 |
| α | 90 [$cm^{-1}$] |
| $\alpha_2$ | $5.5 \times 10^{-4}$ [cm] |
| $\alpha_3$ | $2 \times 10^{-4}$ [cm] |
| $E_p$ | 3.6 [eV] |
| $a_0$ | $-1.033 \times 10^{-4}$ [cm] |
| $a_1$ | $2.7 \times 10^{-10}$ [cm/eV] |
| $a_2$ | $4.33 \times 10^{-17}$ [cm/$eV^2$] |

$$T(t) = \frac{2\exp\left(-\left(\frac{t-t_0}{\sqrt{2}s_{hi}}\right)^2\right)}{\sqrt{2\pi}s_{hi}\sqrt{1+\mathrm{erf}\left(\frac{t_0}{\sqrt{2}s_{hi}}\right)}} \tag{7}$$

Heavy ions possess higher mass and energy than alpha particles, resulting in a significantly greater charge density along the ion track. Consistent with the modeling approach for alpha particles, the parameter values used in the above equations for heavy ion interactions were based on the default silicon material properties provided by Sentaurus TCAD and were directly implemented in the simulation framework of this study [52].

## 3. Results and discussion on radiation effects in BPR-CFET architectures

### 3.1. *Enhanced electric-field distribution in BPR-integrated CFET under radiation exposure*

A comparative analysis of electric-field distributions in Si-CFET structures with BPR and FPR configurations under off-state radiation exposure is presented using cross-sectional views (X-X′, Y-Y′) as shown in Fig. 6. Simulation results indicate that the BPR configuration exhibits increased local electric-field intensity near the silicon substrate due to the proximity between the buried metal rails and the substrate. This enhanced field intensity arises due to the direct interaction between the buried metal rail and the adjacent silicon substrate in the BPR structure. In comparison, the corresponding area in the FPR configuration, consisting of dielectric material, shows a more uniform electric-field distribution with a lower magnitude than that observed in the BPR structure. This amplified electric field significantly increases carrier drift velocities, subsequently enhancing charge collection efficiency and radiation-induced transient current magnitudes.

Fig. 7 highlights how the rail architecture shapes the field direction inside the substrate. In the BPR configuration, the potential applied to the buried metal rail generates strong vertical and lateral components, markedly boosting the net field. This amplification diversifies the drift trajectories and velocities of radiation-induced charge carriers. Conversely, the FPR structure is characterized by a predominantly vertical and spatially uniform electric field. These architectural differences influence charge-collection efficiency and transient current responses during radiation exposure, providing information about the mechanisms of field amplification and increased current density shown in Fig. 7

### 3.2. *Impact of electric-field enhancement on radiation-induced charge collection in BPR-CFETs*

Radiation effects in semiconductor devices fundamentally arise from two primary physical mechanisms: electron-hole pair generation and subsequent drift of charge carriers driven by the internal electric field. When energetic particles interact with semiconductor materials, electrons in the valence band are excited into the conduction band through collisions with lattice atoms, resulting in electron-hole pair formation. This process depends directly on the incident particle energy and the bandgap of the semiconductor material; specifically, in silicon, approximately one electron-hole pair is generated for every 3.6 eV of energy deposited by the radiation particle [54].

The generated charge carriers move by drift under the influence of internal electric fields within the device, which can result in unintended charge collection. Carriers produced in depletion regions, such as drain junctions or well-substrate junctions, are collected quickly due to strong local electric fields, resulting in transient currents in the device. The number of electron-hole pairs formed is directly related to the geometric attributes of the device, particularly the size of its sensitive silicon volume; larger sensitive volumes tend to increase interactions with incident

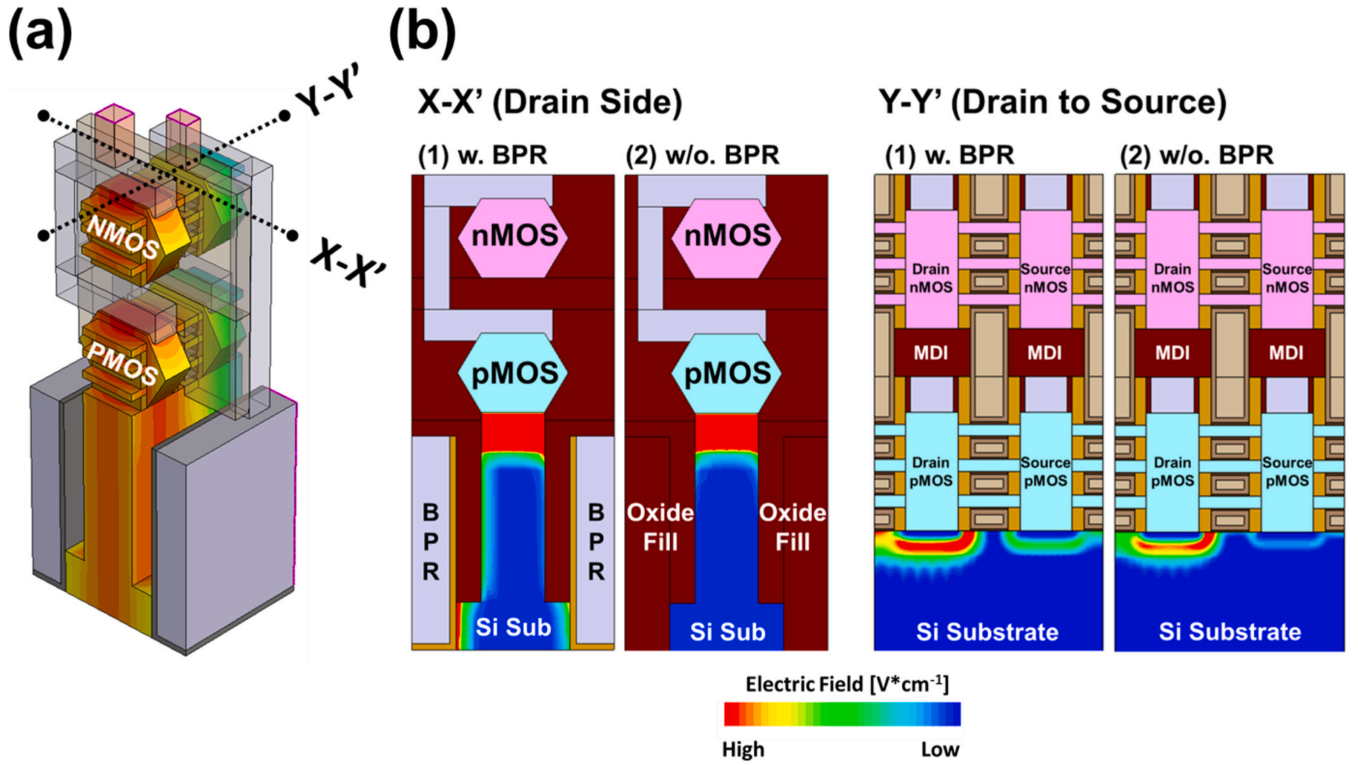


**Fig. 6. Electric field distribution analysis of Si-CFET under radiation particle strike conditions (Vgs=0 V, Vds=Vdd)** (a) 3D device schematic indicating the incident ion track through the vertically stacked pMOS/nMOS pair. (b) X-X'/Y-Y′ cut E-field distribution: BPR Si-CFET exhibits strong vertical confinement, whereas the FPR Si-CFET shows lateral dispersion.

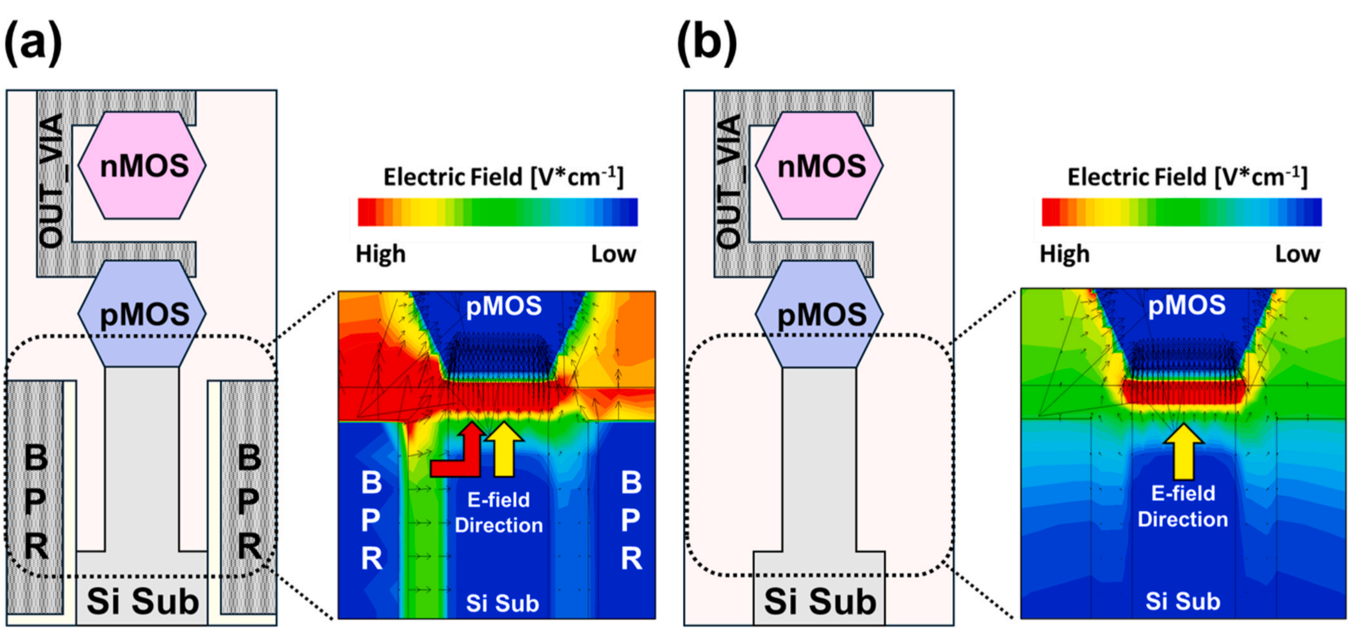


**Fig. 7. Comparative E-field Direction Analysis in Si-CFETs:** (a) E-field distribution and direction in BPR-integrated Si-CFET structure (b) E-field distribution and direction in FPR Si-CFET structure.

radiation particles, leading to more electron-hole pair generation. Carrier drift properties are primarily determined by the distribution, magnitude, and orientation of internal electric fields, which affect drift paths and carrier velocities, and thereby directly impact charge collection efficiency. Stronger electric fields increase carrier drift velocities, lower recombination probability, enable the collection of a greater proportion of generated charges, and yield higher transient currents.

The silicon substrate volume remains constant in both device configurations, with and without the BPR, as analyzed in this study. Consequently, the total number of electron-hole pairs generated by incident radiation is fundamentally the same for each, as shown in Fig. 8 (b). This equivalence arises from the identical likelihood of radiation-particle interaction within equal silicon volumes in both structures. However, as previously discussed, the inclusion of the BPR

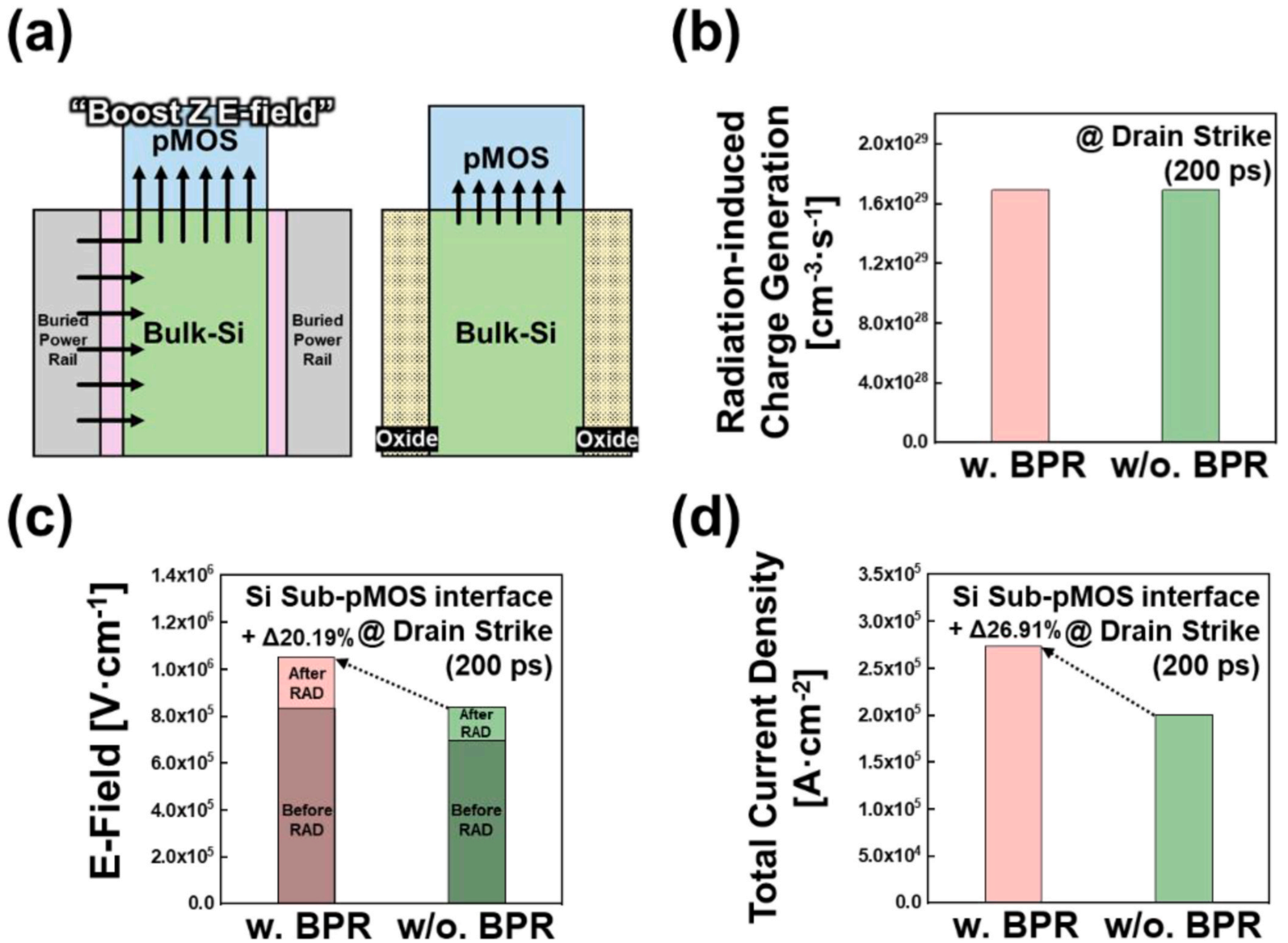


**Fig. 8. Comparison of charge collection characteristics in Si-CFET with and without BPR under identical radiation generation conditions** (a) E-field schematic comparison between w. BPR and w/o. BPR configurations (b) Identical radiation-induced charge generation volume in Si-CFET (c) E-field intensity comparison at Si substrate-pMOS interface (d) Total current density difference due to electric field variation.

fundamentally alters the internal electric field distribution within the silicon substrate. Specifically, the presence of the metal rail adjacent to the substrate introduces adds additional field components that raise the overall strength. This increase in electric field magnitude substantially affects the drift behavior of radiation-induced charge carriers. Due to the enhanced electric fields in the BPR structure, radiation-induced carriers exhibit faster drift velocities, thereby reducing their transit time and significantly lowering the probability of recombination before reaching sensitive device nodes. In the structure incorporating the BPR, the strengthened electric field results in higher drift velocities for these carriers, owing to the direct proportionality between drift velocity and electric field strength. This higher drift velocity results in two notable implications.

First, the shorter transit time lowers recombination probability. Typically, electrons and holes tend to recombine over time, thus diminishing the net charge collected. However, the accelerated drift reduces the likelihood of such recombination events, thereby increasing the quantity of charge successfully collected at the sensitive node. Second, added lateral fields create multiple charge-collection paths. Unlike conventional structures, the BPR configuration establishes horizontal field vectors in addition to vertical ones, enabling charge carriers generated at various spatial locations within the substrate to access multiple routes toward the sensitive nodes more readily. This contributes directly to increased transient current magnitudes under radiation exposure.

In this study, structures with and without BPR have the same silicon volume, and therefore the total pair generation is similar (Fig. 8(b)). However, Fig. 8(c) indicates that the implementation of BPR results in an approximately 20.2 % enhancement in the electric field within the silicon substrate. The stronger field accelerates drift, suppresses recombination, and thus raises collected charge. As shown in Fig. 8(d), the BPR structure's total current density is roughly 26.9 % higher than the conventional FPR structure. Thus, while BPR does not affect pair generation, it notably boosts charge collection and transient current via enhanced electric fields.

### 3.3. Device-level response to single-event strikes

Peak drain current—the maximum transient current after a radiation strike—is a key metric of device-level radiation sensitivity. When a radiation particle interacts with a semiconductor device, it generates electron-hole pairs that are moved by internal electric fields, creating transient current pulses. The magnitude and duration of these pulses influence the potential for changes in the device's logical state. Higher peak drain currents can increase the chances of radiation-induced effects, such as SET and SEU, by exceeding circuit logic thresholds and causing incorrect logic states to propagate.

The magnitude of peak drain currents is fundamentally influenced by the structural attributes of the device. In conventional Si-CFET architectures, the pMOS transistor is fabricated directly on the silicon substrate, while the nMOS transistor is vertically integrated above it [7, 50]. This configuration is typically selected to minimize process complexity and optimize electrical performance, inherently placing the pMOS channel directly adjacent to the silicon substrate.

The structural configuration of the BPR influences the drift characteristics of charge carriers in the silicon substrate. The metal rails within the BPR interact with the silicon substrate, altering the distribution of the local electric field and primarily impacting the nearby pMOS transistor. In contrast, the nMOS transistor, which is located above the pMOS channel, is physically separated from the substrate and thus experiences less direct effect from the changes in the electric field caused by the BPR. The results presented in Fig. 9 quantitatively demonstrate the implications of these structural characteristics on actual radiation sensitivity. As previously depicted in Figs. 6 and 7, the integration of the BPR leads to an approximately 20.2 % increase in local electric field intensity near the silicon substrate. This enhancement is characterized by composite electric field vectors comprising both vertical and lateral components, thereby introducing greater complexity in charge-carrier drift paths and collection trajectories. Furthermore, Fig. 8 provides quantitative evidence that such field augmentation results in an approximate 27 % increase in the radiation-induced current density within the BPR-embedded architecture.

The analysis of isolated nMOS transistor operation, as illustrated in Fig. 9, shows negligible variation between the BPR- and FPR-integrated structures. This outcome is attributed to the spatial detachment of the nMOS device from both the substrate and the BPR, wherein the dominant charge-collection mechanisms are restricted to the upper channel and drain junction. Thus, the enhanced substrate field induced by the BPR has little effect. In contrast, pMOS transistor operation under BPR-integrated conditions demonstrates a peak drain current approximately 29.3 % greater than that observed in the FPR configuration. As previously discussed in Figs. 6–8, this substantial increase is directly attributed to the close physical proximity of the pMOS device to the silicon substrate, whereby BPR-induced amplification of the local electric field significantly alters charge-carrier drift dynamics and promotes efficient collection of radiation-generated charges into the pMOS drain terminal, thereby significantly boosting the transient-current magnitude.

To quantitatively evaluate the structural susceptibility of the BPR-integrated p-channel MOS (pMOS) device under worst-case radiation conditions, transient drain current and collected charge metrics were simulated for a 10 MeV alpha particle strike at the drain terminal at t = 200ps, as depicted in Fig. 10. The transient drain current responses of both BPR and FPR pMOS configurations are presented in Fig. 10(a). Although an off-state bias ideally suppresses drain current, both BPR and FPR structures exhibit transient current pulses due to the generation and rapid collection of radiation-induced electron-hole pairs. However, the intensified local electric field in the BPR device significantly amplifies this response, resulting in a peak drain current that is approximately 41.5 % higher than that of the FPR counterpart, as illustrated in Fig. 10 (a). This observation aligns with previously noted electric field enhancement and directional field diversification effects associated with the BPR structure (Figs. 6–8). Concurrent assessment of the collected charge, defined by

$$Q_{collected} = \int I_{transient}(t)dt$$

The transient charge collection behavior, where $I_{transient}(t)$ denotes the instantaneous transient current, substantiates the previously described trend. As illustrated in Fig. 10(b), the BPR-integrated device exhibits a collected charge that is approximately 42.9 % greater than that of the FPR counterpart. This enhanced charge collection is directly attributable to the elevated transient current profiles driven by the locally intensified electric fields induced by the BPR structure.

The proportional difference between the increases in peak drain current and collected charge is due to their distinct temporal dependencies. In the BPR-integrated structure, the intensified electric field accelerates carrier transport, producing a higher transient current peak, whereas the total collected charge represents the time-integrated response governed by the transient duration, carrier recombination, and post-drift diffusion processes.

Fig. 10 conclusively illustrates that under worst-case alpha particle irradiation conditions, the BPR-integrated pMOS structure exhibits a 41.5 % increase in peak drain current and a 42.9 % enhancement in collected charge relative to the FPR configuration. These quantitative results underscore the pronounced susceptibility of the BPR topology to radiation-induced transients and charge collection phenomena. This

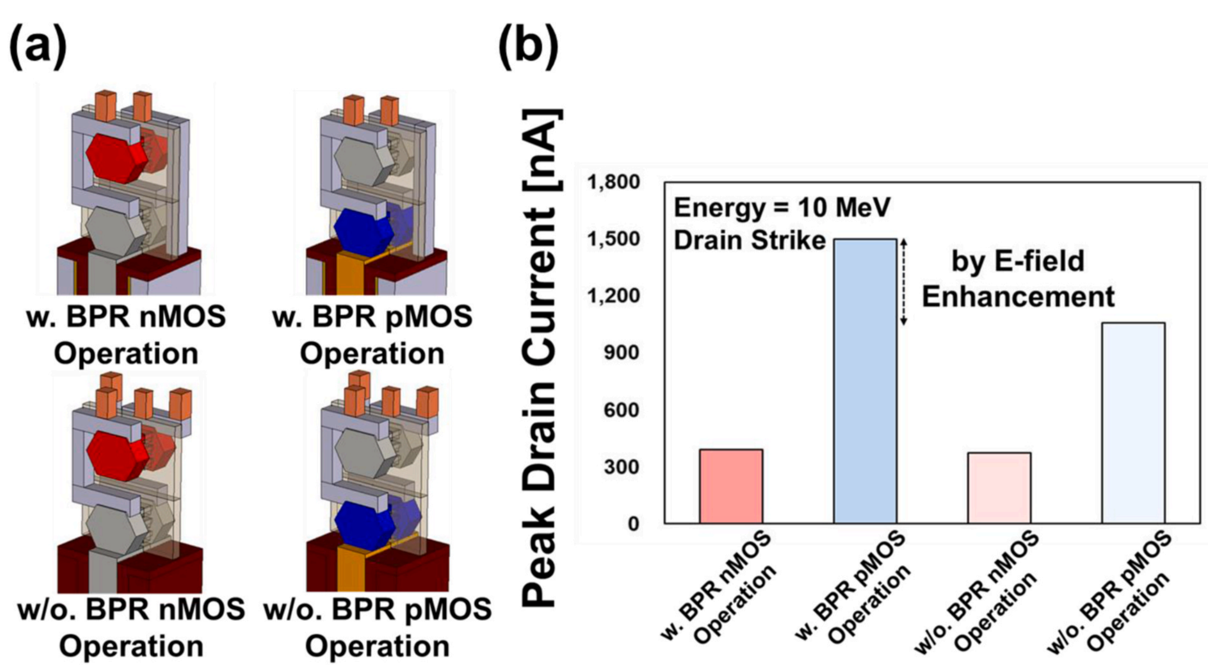


**Fig. 9. Radiation-Induced Peak Drain Current Comparison in Si-CFET Structures with and without BPR** (a) 3D device structures nMOS and pMOS operation modes for w. BPR and w/o. BPR configurations (b) Peak drain current analysis under 10 MeV alpha particle strike demonstrating electric field enhancement effect in BPR structures.

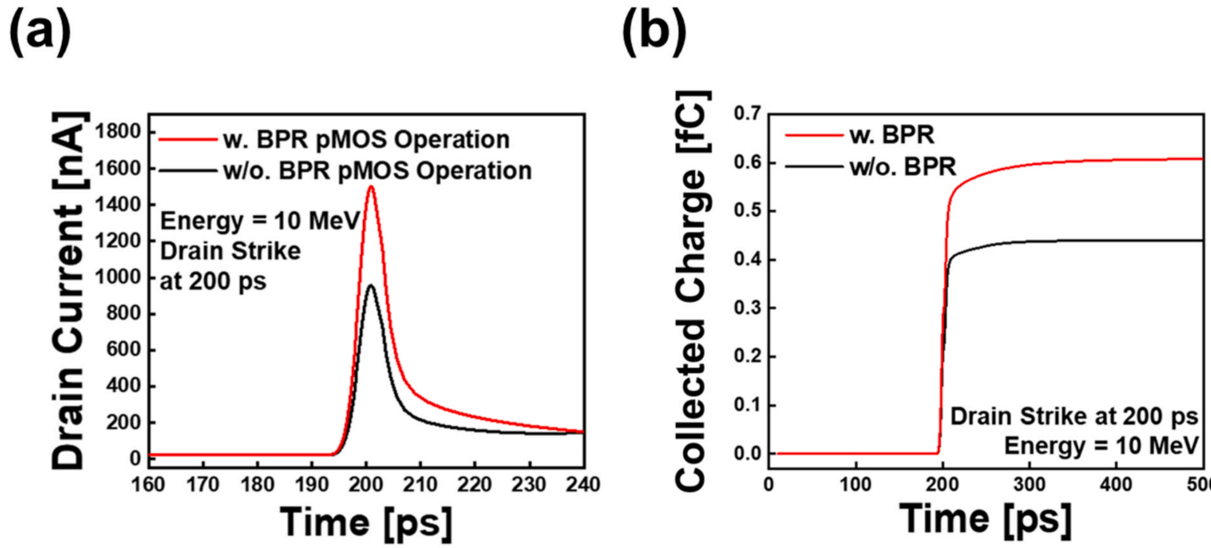


**Fig. 10. Temporal analysis of radiation-induced response in Si-CFET** (a) Drain current transient comparison showing drain current enhancement in BPR structure 10 MeV alpha particle strike at 200 ps. (b) Charge collection showing increased accumulation in w. BPR configuration.

behavior is consistent with previously observed electric field intensification and reconfigured charge collection trajectories discussed in Figs. 6–9.

### 3.4. Circuit-level single-event response of BPR Si-CFET inverters and SRAM

Prior to analyzing circuit-level responses, it is important to note that the adoption of the BPR + CFET architecture in this study stems from its industrial inevitability rather than from any presumed radiation advantage. As the most practical power-delivery solution for sub-3 nm logic, the BPR structure enables reduced IR drop and improved routing efficiency, and enhanced thermal dissipation through buried metal paths, making it an essential integration feature in future nanoscale CMOS technologies. Therefore, evaluating its transient behavior under space-relevant radiation conditions provides critical insight into the reliability boundaries of next-generation logic nodes.

In this study, circuit-level simulations were performed incorporating heavy ion irradiation to emulate realistic spaceborne radiation environments, such as galactic cosmic rays and solar particle events. Unlike electrons or protons, heavy ions have greater mass and charge, thereby creating denser ionization tracks of electron-hole pairs upon impact. This characteristic facilitates effective assessment of memory structures, including static random-access memory (SRAM) flip-flops, under worst-case radiation scenarios. Fig. 11(a) presents the transient response of a complementary field-effect transistor (CFET) inverter subjected to a 1.3 MeV cm$^2$/mg LET heavy ion strike at two distinct time points. At t = 200 ps, with the inverter output in the high state (Vin = 0 V, nMOS off, pMOS on), the resulting voltage excursion was limited to approximately ~0.10 V. Conversely, at t = 600 ps, with the inverter output low (Vin = VDD, nMOS on, pMOS off), the identical ion strike induced a substantial positive voltage excursion of 0.48 V. These outcomes stem from the interplay between BPR-enhanced fields and the pMOS bias state. While the pMOS in its on state provides a low-impedance path to VDD, thereby suppressing voltage perturbations, the off-state configuration yields high output impedance, permitting accumulation of radiation-generated

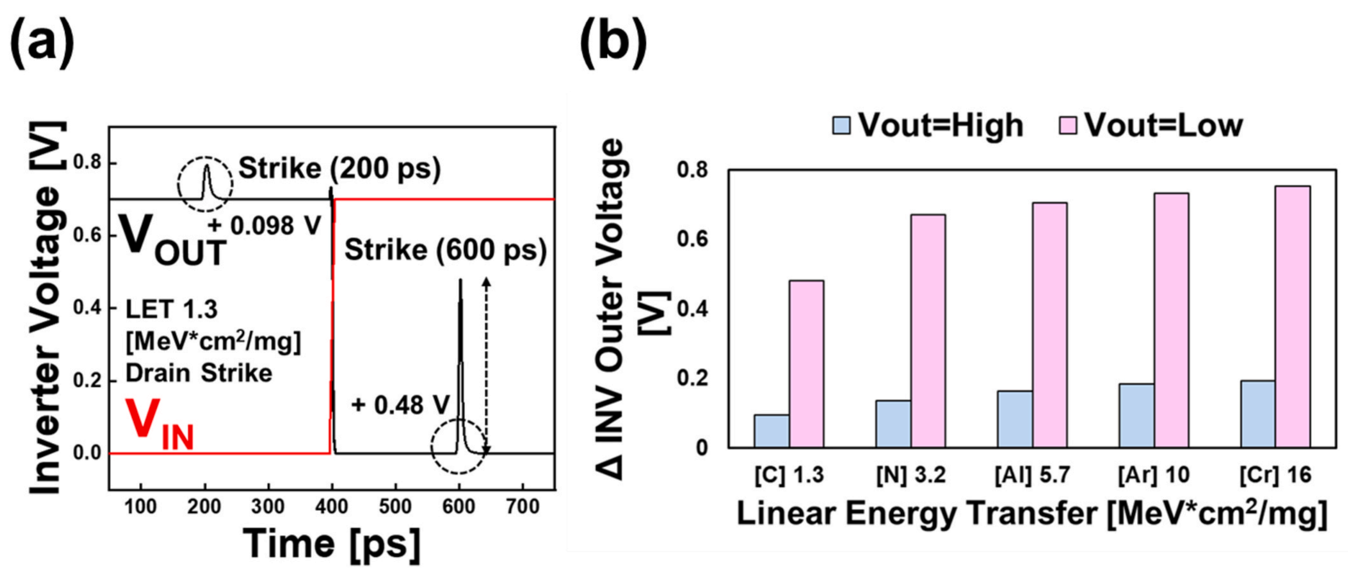


**Fig. 11. Single Event Transient characterization in CFET inverter: Temporal response and LET sensitivity analysis** (a) Inverter voltage transient under heavy ion drain strike conditions (b) Output voltage deviation versus Linear Energy Transfer.

charge at the output node and leading to significant transient voltage deviations.

Fig. 11(b) quantifies the dependence of output-voltage perturbation on LET for various ion species including carbon, nitrogen, aluminum, argon, and chromium over a range of 1.3–16 MeV cm$^2$/mg, based on values reported by the ExoMars project [55]. Experimental data indicate that voltage excursions in the low-state condition exceed those in the high-state by a factor of four to five across all LET levels. Furthermore, both high- and low-state perturbations scale approximately linearly with increasing LET, consistent with the proportional rise in generated charge density. These results demonstrate that CFET inverter radiation sensitivity is highly dependent on device bias state and LET, underscoring the critical need for mitigation strategies such as optimized BPR placement or charge-trapping barriers to ensure reliable logic operation in harsh space environments.

SRAM is widely utilized in aerospace and high-reliability electronic systems due to its high integration density and fast access speed. However, it remains highly vulnerable to SEU in ionizing radiation environments. When high-energy particles such as heavy ions strike the drain region of a memory cell, numerous electron-hole pairs are generated almost instantaneously. If the resultant charge collection exceeds the critical charge, the storage node's logic state inverts, causing a bit flip and potentially leading to system-level failures. With aggressive technology scaling, node capacitance has decreased and critical charge has further reduced, exacerbating SRAM's sensitivity to SEU.

To investigate the flip-flop behavior of the SRAM cell induced by radiation particles, the SEU evaluation was performed under the hold operation mode, where both storage nodes are in a static and metastable state. The hold mode provides a clear and controlled condition for analyzing the intrinsic charge collection and feedback characteristics of the latch without interference from dynamic switching or write operations. This mode is particularly suitable for SEU assessment because any radiation-induced disturbance directly reflects the cell's inherent resilience or vulnerability to soft errors, allowing accurate determination of the critical charge and LET thresholds.

A systematic analysis of the SEU threshold characteristics of Si-CFET SRAM latches—with and without Buried Power Rail (BPR)—is presented in Fig. 12, employing mixed-mode simulation to capture both device- and circuit-level interactions and to quantitatively evaluate the combined influence of LET and supply voltage on the memory cell's radiation tolerance.

The SEU evaluation setup consists of a cross-coupled inverter latch

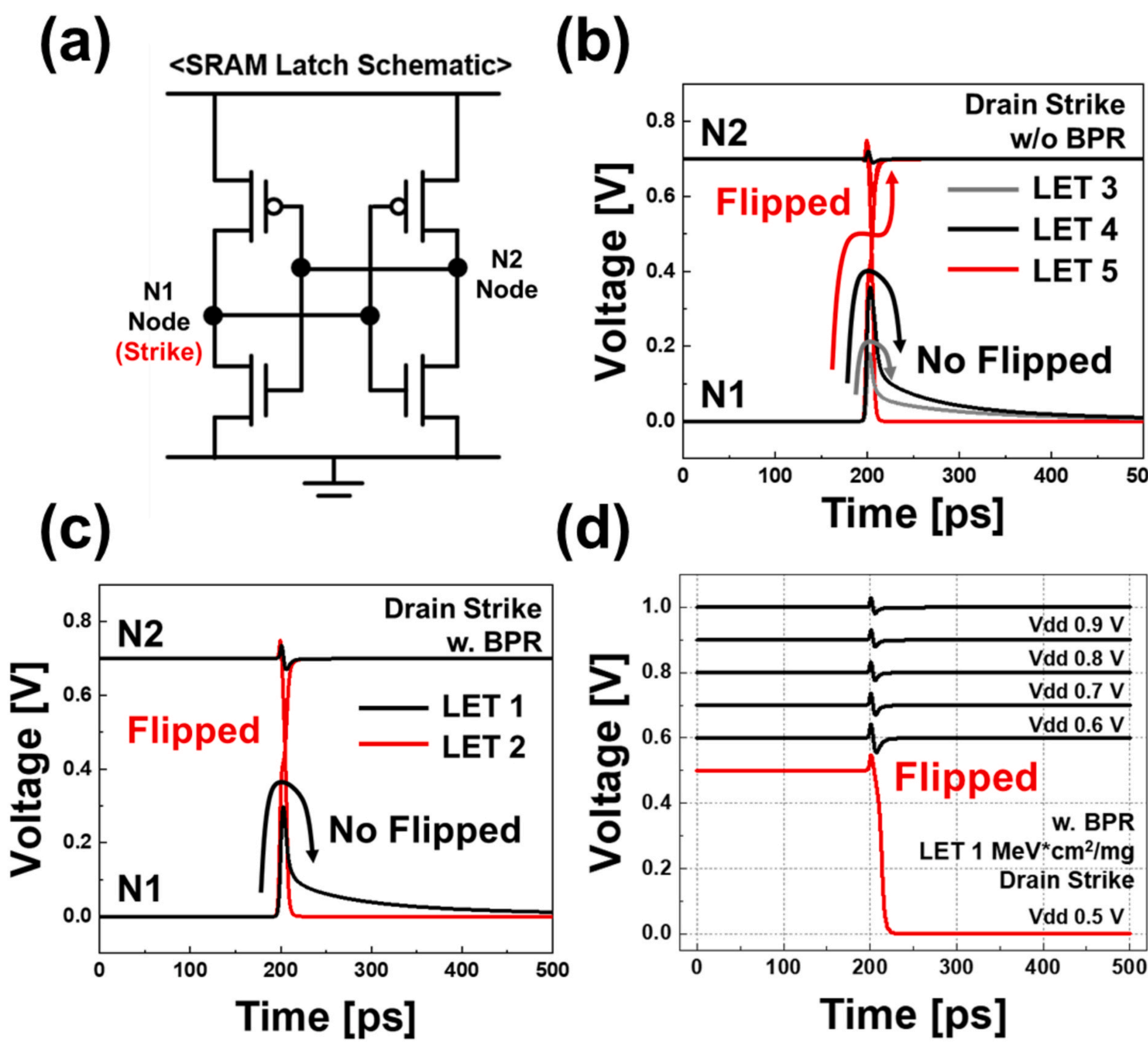


**Fig. 12. Radiation-induced bit flip analysis in BPR Si-CFET SRAM Latch: LET Threshold and Voltage scaling effects** (a) Circuit topology (b) SEU analysis of SRAM latch without BPR (w/o BPR) under LET = 5 MeV cm$^2$/mg drain strike, showing bit-flip occurrence ("Flipped" state). (c) SEU analysis of SRAM latch with BPR (w. BPR) under LET = 2 MeV cm$^2$/mg drain strike, showing bit-flip occurrence ("Flipped" state) (d) Supply-voltage-scaling analysis of the BPR-integrated SRAM latch, showing the critical voltage threshold for SEU occurrence under LET = 1 MeV cm$^2$/mg drain-strike conditions.

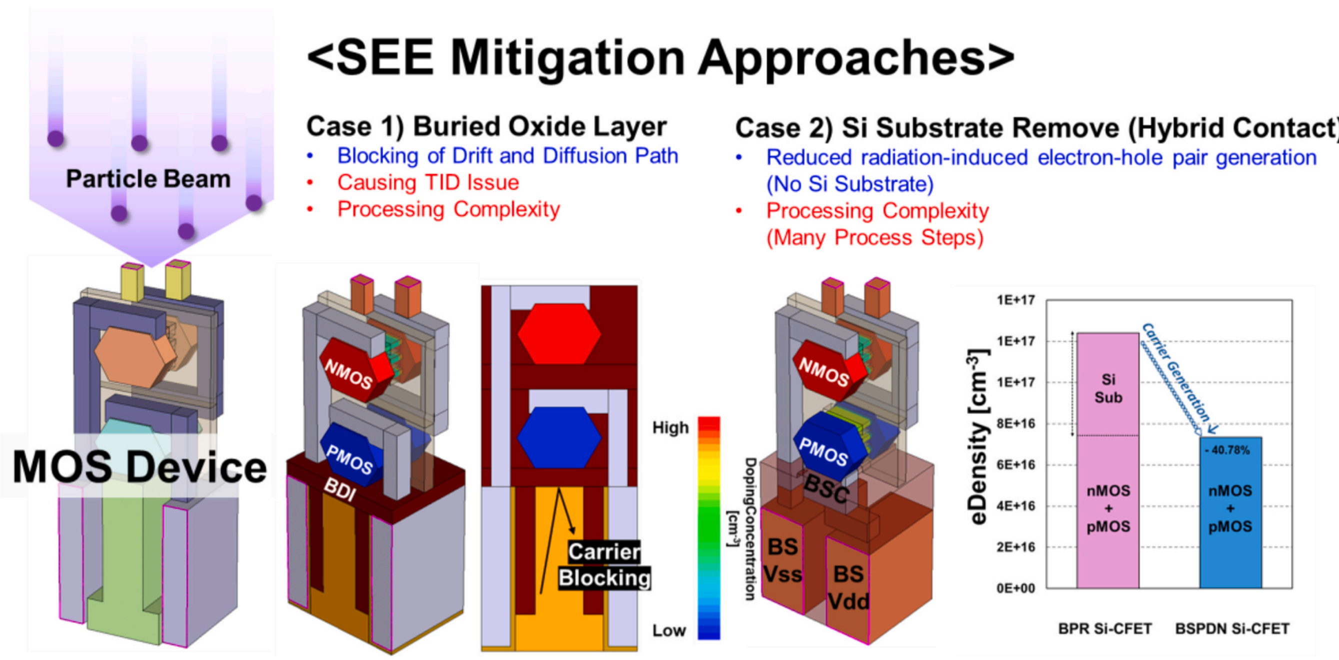


**Fig. 13.** Comparative analysis of SEE mitigation approaches in Si-CFET: Buried Oxide Layer & Si Substrate removal structures (Hybrid Contact).

where storage nodes N1 and N2 are defined, and the radiation strike is applied to the drain of N1, as illustrated in Fig. 12(a). Under an LET condition of 5 MeV cm$^2$/mg, the w/o BPR SRAM latch experiences a clear bit-flip event, as shown in Fig. 12(b). The voltage at node N2 rapidly transitions from '0' to '1', indicating that the collected charge exceeds critical charge and drives the latch into a new stable logic state. This response demonstrates the fundamental SEU mechanism where the deposited charge from a heavy-ion track triggers a regenerative feedback inversion.

In the BPR-integrated SRAM latch, shown in Fig. 12(c), a bit-flip also occurs at a lower LET of 2 MeV cm$^2$/mg. The enhanced electric field induced by the buried metal rail significantly increases carrier drift and charge collection efficiency, resulting in an upset at a much smaller energy deposition compared with the w/o BPR case. This finding confirms that the BPR structure, while advantageous for power delivery and integration, amplifies SEU sensitivity due to intensified substrate coupling and field-assisted charge transport.

Finally, Fig. 12(d) illustrates the supply-voltage-scaling analysis conducted for the BPR-based SRAM latch. Even at a moderate LET of 1 MeV cm$^2$/mg, lowering Vdd increases SEU susceptibility, indicating the deterioration of noise margin under reduced supply conditions. The combined effects of low supply voltage and BPR-induced field enhancement lead to a further decrease in the critical charge threshold, thus narrowing the safe operating voltage range for radiation-hardened SRAM design.

The simulation results presented above indicate that the critical LET required to induce a flip-flop event in the BPR-integrated Si-CFET structure is significantly lower than the radiation hardness levels typically required for space-grade semiconductor devices [56]. This vulnerability is primarily attributed to two interrelated factors: the substantial silicon substrate volume, which increases the generation of electron-hole pairs upon particle incidence, and the strong electric field formed by the close physical proximity between the BPR metal and the silicon substrate, which enhances charge-collection efficiency. The BPR architecture introduces both vertical and lateral electric-field components within the substrate, thereby facilitating rapid drift of radiation-induced carriers toward sensitive nodes. As a result, SEUs and logic state inversions may occur even under relatively low LET conditions, making it difficult to meet the radiation tolerance requirements for aerospace applications. These findings underscore the necessity of structural mitigation strategies to overcome the intrinsic radiation sensitivity associated with BPR-based designs.

### 3.5. Mitigation strategy: Back-Side Contact and buried-oxide isolation

The BPR-CFET structure's vulnerability to low-LET events, due to its strong substrate electric field, makes structural hardening necessary. Although less radiation-tolerant than FPR, BPR offers significant PPA scaling benefits for high-density space systems. Therefore, we propose Buried Dielectric Isolation (BDI) and Back-Side Contact (BSC) as architectural enhancements, not replacements, to maintain BPR's performance and density while mitigating radiation-induced charge collection (see Fig. 13).

BPR integration in Si-CFET SRAMs enhances electric fields due to the close proximity of the buried metal and the substrate. This field enhancement increases charge collection efficiency under radiation, subsequently reducing the critical LET for upset and compromising compliance with space-grade radiation hardness requirements.

To address this critical vulnerability, it is imperative to employ robust structural and layout-level mitigation strategies. This subsection examines two principal countermeasures: the implementation of Buried Dielectric Isolation (BDI) and the adoption of backside contact (BSC) layouts. The following discussion details the operating principles, integration methodologies, as well as the respective advantages and limitations of each approach.

A strategy for mitigation involves placing a buried oxide layer between the BPR and the silicon substrate [57]. This method physically separates the BPR-induced electric field from the substrate, thereby preventing direct coupling and blocking the main drift paths of radiation-generated electron-hole pairs toward sensitive nodes. The dielectric barrier also alters the internal electric-field profile between the BPR and the channel, which can reduce excessive charge collection. While this buried dielectric isolation (BDI) approach can limit SEU effects, it may introduce trade-offs. Insulating materials like $SiO_2$ can increase hole trapping under total ionizing dose (TID) exposure, potentially raising leakage currents over time. Additionally, the dielectric layer may decrease thermal conductivity and increase the complexity of fabrication, impacting reliability and integration processes.

A secondary mitigation strategy entails the physical removal of the silicon substrate. By minimizing the volume of silicon available for particle interaction, this approach substantially decreases the generation of electron-hole pairs, thereby reducing the total charge collected during radiation events. While effectively reducing SEU susceptibility, substrate removal introduces several practical limitations and design challenges. The absence of structural silicon may compromise die mechanical integrity, and altered vertical thermal pathways can adversely affect heat dissipation. Furthermore, increased process complexity and stricter fabrication tolerances present additional challenges to large-scale deployment [59].

A comparative analysis of the BSC Si-CFET structure and the conventional Si-CFET architecture indicates an improvement in SEE tolerance, as shown in Fig. 14. This change reduces both the generation and collection of excess charge, contributing to increased SEE robustness. Previous studies have shown that the buried oxide layer approach (BDI) can improve radiation hardness by blocking charge collection paths through dielectric isolation [57]. In contrast, this work examines a BSC-based solution. Whereas BDI uses insulating layers to mitigate charge transport, the BSC architecture removes the silicon volume associated with particle interactions, aiming to address the source of radiation-induced soft errors.

The process flow for implementing a BSC Si-CFET architecture is depicted in Fig. 14(a) [59]. The procedure initiates with frontside fabrication of the CFET device, followed by wafer-to-wafer bonding onto a carrier substrate. Subsequently, the original silicon substrate is fully removed using a combination of chemical mechanical polishing (CMP) and selective etching techniques. Upon completion of substrate removal, BSCs are established directly from the exposed surface, enabling efficient power delivery and signal access. This methodology not only facilitates backside power-delivery integration and enhances power efficiency and radiation tolerance. Because it shrinks the charge-generation volume and enables vertical routing, the BSC architecture offers a robust, energy-efficient path toward space-grade CFETs.

The output voltage comparison presented in Fig. 14(b) quantitatively demonstrates the superior radiation tolerance of the BSC Si-CFET structure and underscores its strong correlation with the previously identified structural interaction mechanisms between the BPR and the pMOS device. Under an LET condition of 1.3 MeV $cm^2$/mg, a heavy-ion strike at the drain terminal at 600 ps—corresponding to a Vout in the low state (Vin high, nMOS on, pMOS off)—produces a transient voltage perturbation of approximately 0.4 V in the conventional Si-CFET. Conversely, the BSC Si-CFET exhibits a voltage shift of less than 0.1 V under identical conditions. This notable reduction in voltage perturbation confirms that the removal of the silicon substrate effectively suppresses electric field enhancement induced by the BPR, thereby reducing radiation-induced charge collection and mitigating the occurrence of SETs.

A comprehensive LET analysis of the BSC SRAM latch, illustrated in Fig. 14(c), offers a systematic assessment of radiation tolerance at the memory level. Simulations performed over an LET range of 1–34 MeV $cm^2$/mg indicate that the BSC-based structure remains stable up to an LET of 33, with bit flips only occurring at LET values of 34 or higher. This outcome signifies a marked enhancement in SEU robustness compared to conventional Si-CFET SRAM, where upset events are detected at LETs as low as 2 MeV $cm^2$/mg. The observed 17-fold increase in critical LET underscores the efficacy of silicon substrate removal in mitigating radiation-induced charge collection and establishes the BSC architecture as a robust solution for space-grade memory applications.

Radiation hardening in BSC stems from three factors: (i) *Volume reduction* — substrate removal lowers the EHP generation volume; (ii) *Field relaxation* — erasing the BPR-substrate field cuts charge-collection efficiency; (iii) *Path suppression* — absent substrate blocks diffusive transport to sensitive nodes."

While the BSC architecture offers enhanced radiation tolerance, it also presents several technical considerations. First, process complexity

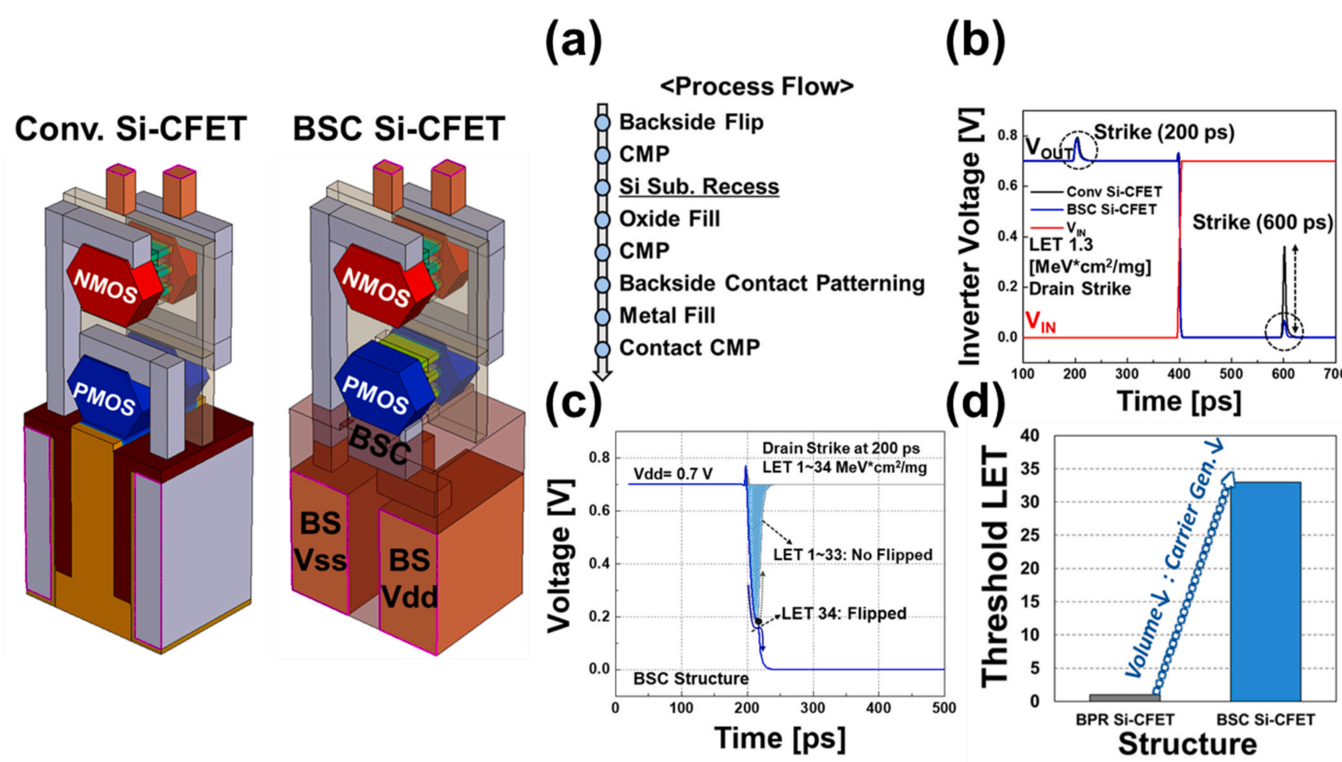


**Fig. 14. BSC (Back Side Contact) structure integration for radiation hardness enhancement** (a) Fabrication process (b) Voltage transfer characteristics of conventional Si-CFET vs BSC Si-CFET inverters (c) LET-dependent flip behavior in BSC structures (d) Threshold LET improvement comparison.

increases as implementation involves additional steps such as wafer thinning and backside lithography, which can raise fabrication costs and integration challenges [58–60]. Second, mechanical stability may be affected due to the complete removal of the silicon substrate, potentially reducing wafer rigidity and increasing susceptibility to mechanical damage during processing and handling. Third, thermal management requires attention because modifications in heat dissipation pathways under the BSC configuration necessitate adjustments to the thermal design to support reliable operation in high-power or demanding environments [60].

In summary, the BSC Si-CFET architecture utilizes complete removal of the silicon substrate to improve radiation tolerance by reducing the volume available for radiation-induced charge generation. This design addresses reliability requirements relevant to space-grade semiconductor applications. The enhanced durability of BSC-based designs is applicable to space missions where operation under extreme environmental conditions is required.

## 4. Conclusion

This study quantifies single-event effects in buried-power-rail (BPR) nanosheet CFETs. Three-dimensional TCAD and mixed-mode circuit simulations were used. The results demonstrate that BPR integration enhances the internal electric field by approximately 20.2 %, which in turn boosts transient current by up to 41.5 % than those observed in conventional front power rail (FPR) architectures. This amplification significantly increases the susceptibility of the device to radiation-induced disturbances. Circuit-level analyses further reveal that SRAM latch circuits employing BPR integration exhibit increased vulnerability to bit-flip events at relatively low Linear Energy Transfer (LET) levels (1–2 MeV $cm^2$/mg), with this sensitivity being further exacerbated at reduced supply voltages. These findings highlight critical challenges in designing low-power, radiation-hardened systems. To address these issues, structural hardening techniques such as buried dielectric isolation and backside contact were proposed and shown to raise the critical LET from 2 to 34 MeV $cm^2$/mg. The insights gained from this work offer practical design strategies for improving radiation resilience in ultra-scaled logic and memory devices intended for spaceborne and other high-reliability applications.

## CRediT authorship contribution statement

**Dongwook Kim:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hanggyo Jung:** Writing – review & editing, Project administration, Formal analysis, Conceptualization. **Jimyeong Lee:** Project administration, Data curation. **Seungkyu Kim:** Project administration, Data curation. **Jongwook Jeon:** Writing – review & editing, Supervision, Software, Resources, Funding acquisition, Formal analysis, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## References

[1] Ali Razavieh, Peter Zeitzoff, Edward J. Nowak, Challenges and limitations of CMOS scaling for FinFET and beyond architectures, IEEE Trans. Nanotechnol. 18 (2019) 999–1004.
[2] Chih-Hao Chang, et al., Critical process features enabling aggressive contacted gate pitch scaling for 3nm CMOS technology and beyond. 2022 International Electron Devices Meeting (IEDM), IEEE, 2022.
[3] Krishna K. Bhuwalka, et al., Optimization and benchmarking FinFETs and GAA nanosheet architectures at 3-nm technology node: impact of unique boosters, IEEE Trans. Electron. Dev. 69 (8) (2022) 4088–4094.
[4] Robert H. Dennard, et al., Design of ion-implanted MOSFETs with very small physical dimensions, IEEE J. Solid State Circ. 9 (5) (2003) 256–268.
[5] Isabelle Ferain, Cynthia A. Colinge, Jean-Pierre Colinge, Multigate transistors as the future of classical metal–oxide–semiconductor field-effect transistors, Nature 479 (7373) (2011) 310–316.
[6] Junyeol Lee, et al., Exploring optimal TMDC multi-channel GAA-FET architectures at sub-1nm nodes, J. Sci. Adv. Mater. Devices (2025) 100931.
[7] Julien Ryckaert, et al., The Complementary FET (CFET) for CMOS scaling beyond N3. 2018 IEEE Symposium on VLSI Technology, IEEE, 2018.
[8] Jiawei Li, et al., Performance comparison of vertically stacked nanosheet CFET and standard CMOS without and with parasitic channels, Microelectron. Eng. 279 (2023) 112053.
[9] S. Subramanian, et al., First monolithic integration of 3D complementary FET (CFET) on 300mm wafers. 2020 Ieee Symposium on Vlsi Technology, IEEE, 2020.
[10] C.-Y. Huang, et al., 3-D self-aligned stacked NMOS-on-PMOS nanoribbon transistors for continued Moore's law scaling. 2020 IEEE International Electron Devices Meeting (IEDM), IEEE, 2020.
[11] Seung Kyu Kim, et al., A comparative analysis of middle-of-line contact architectures for complementary FETs, IEEE Access (2024).
[12] Xiaoqiao Yang, et al., Compact modeling of process variations in nanosheet complementary FET (CFET) and circuit performance predictions, IEEE Trans. Electron. Dev. 70 (7) (2023) 3935–3942.
[13] Julien Ryckaert, et al., Extending the roadmap beyond 3nm through system scaling boosters: a case study on Buried Power Rail and Backside Power Delivery. 2019 Electron Devices Technology and Manufacturing Conference (EDTM), IEEE, 2019.
[14] S.C. Song, et al., System design technology co-optimization for 3D integration at< 5nm nodes. 2021 IEEE International Electron Devices Meeting (IEDM), IEEE, 2021.
[15] G. Hellings, G. Hiblot, J. Ryckaert, Process architectures changes to improve power delivery, in: IEDM Tech. Dig., Dec. 2022, pp. 1–78.
[16] Sandra Maria Shaji, et al., A comparative study on front-side, buried and back-side power rail topologies in 3nm technology node. 2023 IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED), IEEE, 2023.
[17] Daniel Gall, Atharv Jog, Tianji Zhou, Narrow interconnects: the most conductive metals. 2020 IEEE International Electron Devices Meeting (IEDM), IEEE, 2020.
[18] Takeshi Nogami, et al., Comparison of key fine-line BEOL metallization schemes for beyond 7 nm node. 2017 Symposium on VLSI Technology, IEEE, 2017.
[19] R. Chen, et al., Power, performance, area, and thermal analysis of 2D and 3D ICs at A14 node designed with back-side power delivery network. 2022 International Electron Devices Meeting (IEDM), IEEE, 2022.
[20] Md Obaidul Hossen, et al., Power delivery network (PDN) modeling for backside-PDN configurations with buried power rails and μ TSVs, IEEE Trans. Electron. Dev. 67 (1) (2019) 11–17.
[21] Divya Prasad, et al., Buried power rails and back-side power grids: arm® CPU power delivery network design beyond 5nm. 2019 IEEE International Electron Devices Meeting (IEDM), IEEE, 2019.
[22] Anshul Gupta, et al., High-aspect-ratio ruthenium lines for buried power rail. 2018 IEEE International Interconnect Technology Conference (IITC), IEEE, 2018.
[23] Eunbin Park, Taigon Song, Complementary FET (CFET) standard cell design for low parasitics and its impact on VLSI prediction at 3-nm process, IEEE Trans. Very Large Scale Integr. Syst. 31 (2) (2022) 177–187.
[24] A. Gupta, et al., Buried power rail scaling and metal assessment for the 3 nm node and beyond. 2020 IEEE International Electron Devices Meeting (IEDM), IEEE, 2020.
[25] Rahul Mathur, et al., Buried bitline for sub-5nm sram design. 2020 IEEE International Electron Devices Meeting (IEDM), IEEE, 2020.
[26] Li Yonghong, et al., Atmospheric neutron inducing single event effects on AI chips manufacturing with 8 nm FinFET, Nucl. Eng. Technol. 57 (2025) 11.
[27] Eunju Jo, et al., Radiation tolerant capacitor-SRAM without area overhead, Nucl. Eng. Technol. 56 (8) (2024) 2916–2922.
[28] T.S. Nidhin, et al., Understanding radiation effects in SRAM-based field programmable gate arrays for implementing instrumentation and control systems of nuclear power plants, Nucl. Eng. Technol. 49 (8) (2017) 1589–1599.
[29] Taiki Uemura, Yoshiharu Tosaka, Shigeo Satoh, Neutron-induced soft-error simulation technology for logic circuits, Jpn. J. Appl. Phys. 45 (4S) (2006) 3256.
[30] Joseph R. Srour, J. Marshall Cheryl, W. Marshall Paul, Review of displacement damage effects in silicon devices, IEEE Trans. Nucl. Sci. 50 (3) (2003) 653–670.
[31] D.R. Hughart, et al., Mechanisms of interface trap buildup and annealing during elevated temperature irradiation, IEEE Trans. Nucl. Sci. 58 (6) (2011) 2930–2936.
[32] Paul E. Dodd, W. Massengill Lloyd, Basic mechanisms and modeling of single-event upset in digital microelectronics, IEEE Trans. Nucl. Sci. 50 (3) (2003) 583–602.
[33] Yoni Xiong, et al., Soft error rate predictions for terrestrial neutrons at the 3-nm bulk FinFET technology. 2023 IEEE International Reliability Physics Symposium (IRPS), IEEE, 2023.

[34] Patrick Nsengiyumva, et al., A comparison of the SEU response of planar and FinFET D flip-flops at advanced technology nodes, IEEE Trans. Nucl. Sci. 63 (1) (2016) 266–272.
[35] Taiki Uemura, et al., Investigation of alpha-induced single event transient (SET) in 10 nm FinFET logic circuit. 2018 IEEE International Reliability Physics Symposium (IRPS), IEEE, 2018.
[36] Zhengxin Zhang, et al., Single event upset and radiation hardening of the complementary FET (CFET) based 6T-SRAM. 2024 2nd International Symposium of Electronics Design Automation (ISEDA), IEEE, 2024.
[37] Gunhee Choi, Seungkyu Kim, Jongwook Jeon, Investigation of radiation effects on Multichannel Nanosheet-FETs, Forksheet-FETs, and Complementary-FETs, IEEE Trans. Electron. Dev. (2024).
[38] V. Vibhu, Sparsh Mittal, Vivek Kumar, Machine learning-based model for single event upset Current prediction in 14nm FinFETs. 2023 36th International Conference on VLSI Design and 2023 22nd International Conference on Embedded Systems (VLSID), IEEE, 2023.
[39] Jinsu Jeong, et al., Novel trench inner-spacer scheme to eliminate parasitic bottom transistors in silicon nanosheet FETs, IEEE Trans. Electron. Dev. 70 (2) (2022) 396–401.
[40] Yeji Lee, et al., Investigation on the effects of interconnect RC in 3nm technology node using path-finding process design kit, IEEE Access 10 (2022) 80695–80702.
[41] Hao Yu, et al., Titanium (germano-) silicides featuring 10− 9 Ω· Cm 2 contact resistivity and improved compatibility to advanced CMOS technology. 2018 18th International Workshop on Junction Technology (IWJT), IEEE, 2018.
[42] Pieter Schuddinck, et al., Device-, circuit-& block-level evaluation of CFET in a 4 track library. 2019 Symposium on VLSI Technology, IEEE, 2019.
[43] N. Horiguchi, et al., 3D stacked devices and MOL innovations for post-nanosheet CMOS scaling. 2023 International Electron Devices Meeting (IEDM), IEEE, 2023.
[44] Julien Ryckaert, et al., Extending the roadmap beyond 3nm through system scaling boosters: a case study on Buried Power Rail and Backside Power Delivery. 2019 Electron Devices Technology and Manufacturing Conference (EDTM), IEEE, 2019.
[45] Dmitry Yakimets, et al., Inflection points in GAA NS-FET to C-FET scaling considering impact of DTCO boosters, IEEE Trans. Electron. Dev. 71 (4) (2024) 2309–2314.
[46] Jun-Sik Yoon, et al., Metal source-/drain-induced performance boosting of sub-7-nm node nanosheet FETs, IEEE Trans. Electron. Dev. 66 (4) (2019) 1868–1873.
[47] Marko Radosavljević, et al., Demonstration of a stacked CMOS inverter at 60nm gate pitch with power via and direct backside device contacts. 2023 International Electron Devices Meeting (IEDM), IEEE, 2023.
[48] Jaehyun Park, et al., Highly manufacturable self-aligned direct backside contact (SA-DBC) and backside gate contact (BGC) for 3-dimensional stacked FET at 48nm gate pitch. 2024 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits), IEEE, 2024.
[49] S. Liao, et al., Complementary field-effect transistor (CFET) demonstration at 48nm gate pitch for future logic technology scaling. 2023 International Electron Devices Meeting (IEDM), IEEE, 2023.
[50] N. Loubet, et al., Stacked nanosheet gate-all-around transistor to enable scaling beyond FinFET. 2017 Symposium on VLSI Technology, IEEE, 2017.
[51] Harold Hughes, et al., Total ionizing dose radiation effects on 14 nm FinFET and SOI UTBB technologies. 2015 IEEE Radiation Effects Data Workshop (REDW), IEEE, 2015.
[52] Sentaurus User's Manual, Synopsys, Inc., Mountain View, CA, 2021.
[53] A. Erlebach, Modellierung Und Simulation Strahlensensitiver Halbleiterbauelemente, Aachen, Shaker, 1999.
[54] R.C. Alig, S. Bloom, Electron-hole-pair creation energies in semiconductors, Phys. Rev. Lett. 35 (22) (1975) 1522.
[55] P. Manzano, et al., Heavy ion latch-up test on dsPIC microcontroller to be used in ExoMars 2020 mission. 2017 IEEE Radiation Effects Data Workshop (REDW), IEEE, 2017.
[56] K.A. LaBel, Single event effects (SEEs) and requirements, HEART Short Course (March 12, 2003). Albuquerque, NM.
[57] Gunhee Choi, Jongwook Jeon, Radiation effects on multi-channel Forksheet-FET and Nanosheet-FET considering the bottom dielectric isolation scheme, Nucl. Eng. Technol. 56 (11) (2024) 4679–4687.
[58] P. Zhao, et al., Proceedings of the 2024 Symposium on VLSI Technology and Circuits (VLSI), IMEC, 2024.
[59] S. Yang, et al., Proceedings of the 2023 Symposium on VLSI Technology and Circuits (VLSI), 2023 imec.
[60] R. Chen, et al., Proceedings of the 2022 IEEE International Electron Devices Meeting (IEDM), 2022 imec.